

Example 27: Display the graph of the function $f(x) = x^2$ from the set of integers to the set of integers.

Solution:

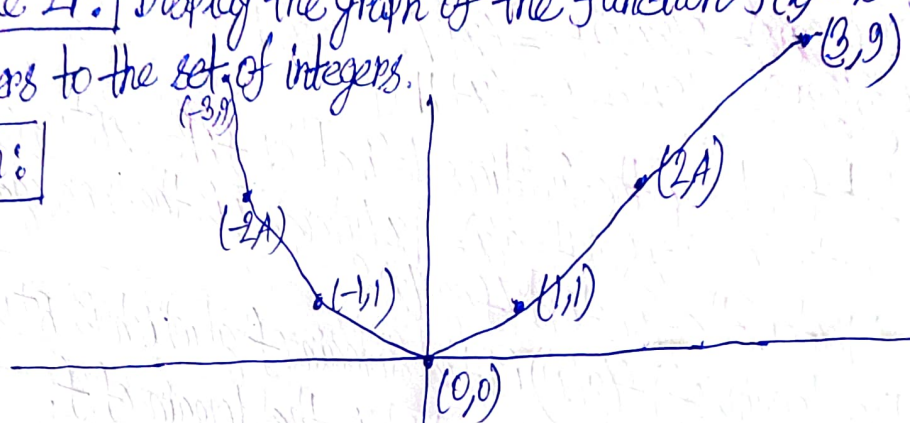


FIGURE 9: The graph of $f(x) = x^2$ from $\mathbb{Z}$ to $\mathbb{Z}$

2.3.5 Some Important Functions

Definition 12: The **floor function** assigns to the real number x the largest integer that is less than or equal to x . The value of the floor function at x is denoted by $\lfloor x \rfloor$. The **ceiling function** assigns to the real number x the smallest integer that is greater than or equal to x . The value of the ceiling function at x is denoted by $\lceil x \rceil$.

Remark: The floor function is often also called the **greatest integer function**. It is often denoted by $[x]$.

These functions are often used when objects are counted. They play an important role in the analysis of the number of steps used by procedures to solve problems of a particular size.

Example 28: These are some of the values of the floor and ceiling functions

$$\lfloor \frac{1}{2} \rfloor = 0, \lceil \frac{1}{2} \rceil = 1, \lfloor -\frac{1}{2} \rfloor = -1, \lceil -\frac{1}{2} \rceil = 0, \lfloor 3.1 \rfloor = 3, \lceil 3.1 \rceil = 4, \lfloor 7 \rfloor = 7, \lceil 7 \rceil = 7. \square$$

